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Ques. What is Laravel?

- Laravel is a free and open-source **PHP web application framework** used to build modern, secure, and scalable web applications quickly.
- It follows the **MVC (Model-View-Controller)** architecture pattern, which helps organize code and makes applications easier to maintain.
- It is created by **Taylor Otwell**
- The first version of laravel is released on 9 June 2011.
- the latest stable Laravel version is Laravel **13**, released on **March 17, 2026**. It requires PHP 8.3 or higher.

What is the Features of Laravel?

1. MVC Architecture – Separates business logic, UI, and data.
 2. Eloquent ORM – Easy database operations using PHP models.
 3. Routing – Clean and simple URL management.
 4. Blade Templating Engine – Dynamic and reusable views.
 5. Authentication & Authorization – Built-in login and access control.
 6. Middleware – Filter HTTP requests.
 7. Migration & Seeder – Manage database schema and sample data.
 8. Artisan CLI – Command-line tool for development tasks.
 9. Queue System – Handle background jobs.
 10. Security – Protection against SQL Injection, CSRF, XSS, etc.
- Inbuilt CRSF (cross-site request forgery) Protection.
 - Inbuilt paginations
 - middleware
 - Query builder available
 - Reverse Routing
 - Restful Controllers
 - Autamatic Pagination
 - Unit Testing
 - Homestead
 - Route caching
 - IOC (Inverse of Control) Container Or service container.

Why Use Laravel?

- Laravel is used to build web applications, REST APIs, e-commerce websites, CRM systems, ERP systems, and enterprise applications quickly and securely.
- Main Reasons to Use Laravel
 - **Rapid Development**:- Many features are built-in, reducing development time.
 - **MVC Architecture**:- Keeps code organized and easy to maintain.
 - **Database Management**:- Eloquent ORM simplifies database operations.
 - **Authentication & Authorization**:- Login, registration, roles, and permissions can be implemented easily.
 - **Security**:- Protection against SQL Injection, CSRF, and XSS attacks.
 - **REST API Development**:- Easy creation of APIs for mobile and web applications.

- Artisan CLI:- Generates controllers, models, migrations, and more with simple commands.
- Caching & Queues:- Improves application performance.
- Large Community Support:- Extensive documentation and packages available.
- Faster development
- Clean code structure
- Large community support

What are pros and cons of using Laravel Framework?

Pros of using Laravel Framework

1. Laravel framework has in-built lightweight blade template engine to speed up compiling task and create layouts with dynamic content easily.
2. Hassles code reusability.
3. Eloquent ORM with PHP active record implementation
4. Built in command line tool "Artisan" for creating a code skeleton , database structure and build their migration

Cons of using laravel Framework

1. Development process requires you to work with standards and should have real understanding of programming
2. Laravel is new framework and composer is not so strong in compare to npm (for node.js), ruby gems and python pip.
3. Development in laravel is not so fast in compare to ruby on rails.
4. Laravel is lightweight so it has less inbuilt support in compare to django and rails. But this problem can be solved by integrating third party tools, but for large and very custom websites it may be a tedious task.

Ques. What are available databases supported by Laravel?

- PostgreSQL
- SQL Server
- SQLite
- MySQL

Ques. What are the steps to install Laravel with composer?

Laravel installation steps:-

- Download composer from <https://getcomposer.org/download> (if you don't have a composer on your system)
- Open cmd
- Goto your htdocs folder.

```
C:\xampp\htdocs> composer create-project laravel/laravel projectname
```

- If you install some particular version, then you can use

```
composer create-project laravel/laravel project_name "8.0"
```

- If you did not mention any particular version, then it will install with the latest version.

Ques. What is the templating engine used in Laravel?

- The templating engine used in Laravel is **Blade**.
- **Displaying data** If you want to print the value of a variable, then you can do so by simply enclosing the variable within the **curly brackets**.
- **Syntax:-**

```
{{ $variable }}
```

Ques. How to put Laravel applications in maintenance mode?

Laravel applications can be put into maintenance mode using the below command:

```
php artisan down
```

And can put the application again on live using the below command:

```
php artisan up
```

Ques. List some Aggregates methods provided by query builder in Laravel?

- count()
- max()
- min()
- avg()
- sum()

How to check laravel version?

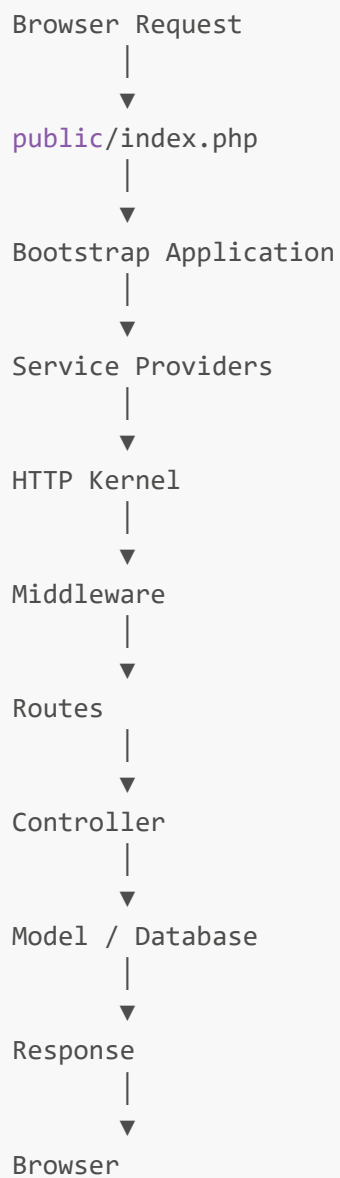
```
# bash
php artisan --version
php artisan -V

# From within the Application Code
echo app()->version();
```

```
# In a Blade template  
{{ app()->version() }}
```

Explain Laravel's request lifecycle.

- In Laravel, a request enters through `public/index.php`, is processed by the HTTP Kernel and middleware, matched to a route, handled by a controller, interacts with models if needed, and then returns a response to the browser.
- Step-by-Step Explanation
 1. Request Enters **public/index.php**
 - This is the **entry point** of every Laravel application.
 - It loads Composer autoload files and starts Laravel.
 2. Bootstrap the Application
 - Laravel loads configuration files.
 - Environment variables (**.env**) are loaded.
 - Service providers are registered.
 3. HTTP Kernel Handles Request
 - The request is passed to the HTTP Kernel.
 - The kernel manages middleware execution.
 4. Middleware Execution
 - Middleware checks or modifies requests before they reach controllers.
 - Examples:
 - Authentication
 - CSRF Protection
 - CORS
 - Logging
 5. Route Matching
 - Laravel finds the matching route.
 6. Controller Execution
 - The controller method is executed.
 7. Model & Database Interaction
 - Laravel uses Eloquent ORM to fetch or save data.
 8. Response Generation
 - The controller returns:
 - View
 - JSON
 - Redirect
 - File Download



What is Artisan?

- **Artisan** is **Laravel's command-line tool** used to automate development tasks such as creating controllers, models, migrations, running migrations, clearing cache, and executing custom commands.

Command	Purpose
php artisan serve	Start Laravel development server
php artisan make:controller UserController	Create a controller
php artisan make:model User	Create a model
php artisan make:migration create_users_table	Create a migration
php artisan migrate	Run migrations
php artisan db:seed	Run seeders
php artisan route:list	Show all routes
php artisan cache:clear	Clear application cache
php artisan config:clear	Clear config cache

What is Composer?

- Composer is a **dependency management tool for PHP** that helps developers install and manage third-party libraries and project dependencies automatically.

Why Use Composer?

- Install PHP packages easily
- Manage project dependencies
- Update libraries with a single command
- Autoload classes automatically
- Maintain consistent project versions

Example

- Install project dependencies:

```
composer install
```

- Add a new package:

```
composer require laravel/sanctum
```

- Update dependencies:

```
composer update
```

What is Service Container?

- The Service Container is Laravel's **Dependency Injection (DI) container**. It is responsible for managing class dependencies and **automatically injecting them when needed**.

Why Use Service Container?

- Reduces manual object creation
- Supports Dependency Injection
- Makes code loosely coupled
- Improves testing and maintainability